# Slippers2Sat-2 (S2S): Digipeater Mission

S2S-2 Team

Kathmandu, Nepal

# MSN1: Digipeater Mission (DP-M)

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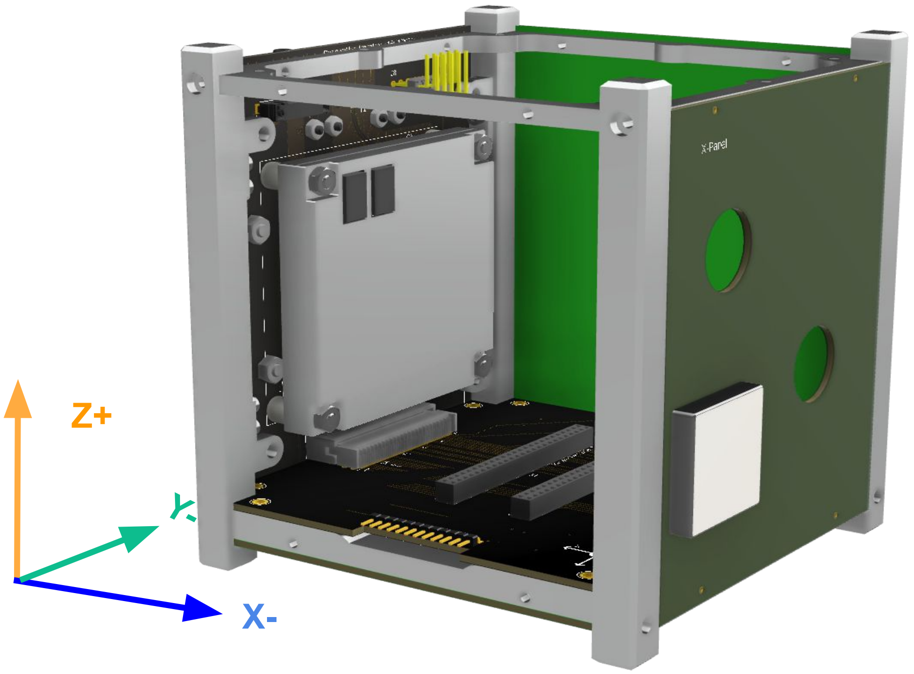
# Digipeater (DP-M)

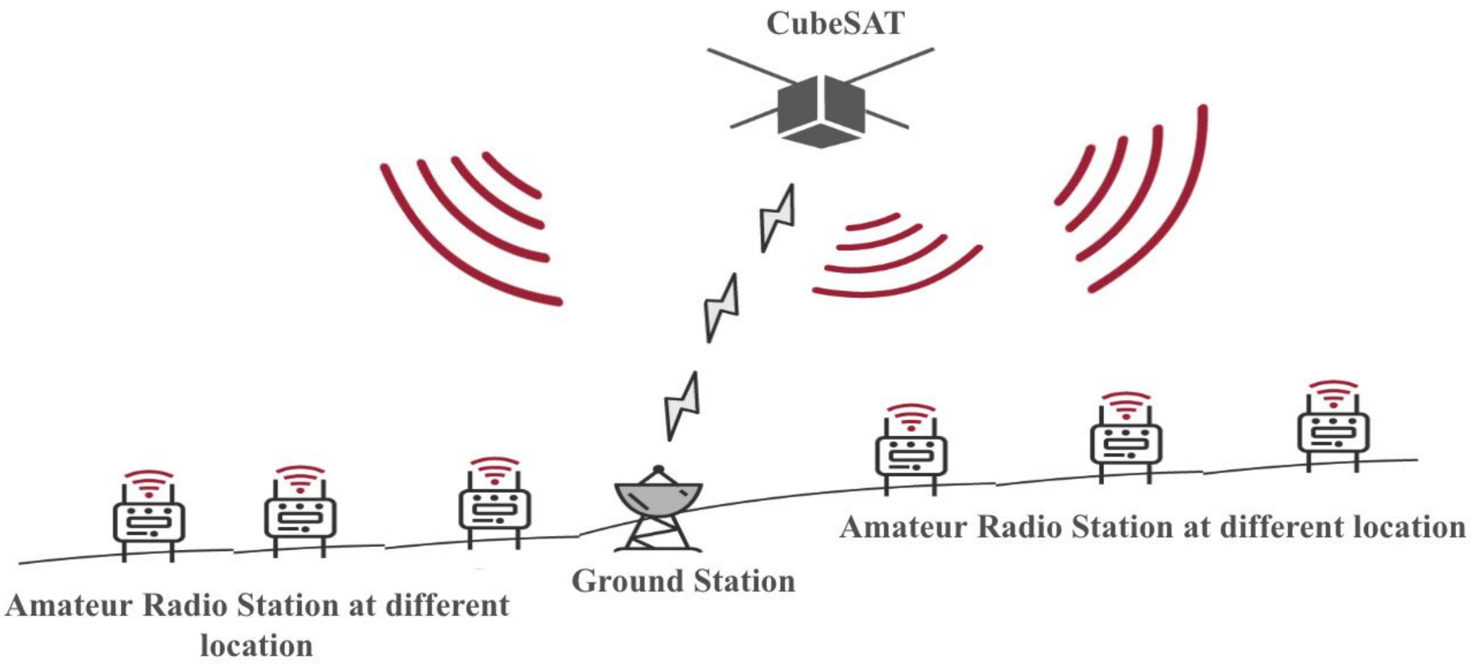


# Introduction

- **Digipeater (Digital Repeater):**
    - A special type of repeater used in Amateur Radio for digital communication.
    - AX.25 protocol is most common communication protocol use in CubeSat
    - However, the packet format may not be required to be AX.25 packet for digipeating
    - Communicates in Half-duplex mode
    - Facilitates store and forward operation

In Slipper2Sat, Software Digipeater is used in **UHF band**

# Introduction



# Statement and Objective

- **<u>Mission Statement</u>**
  - When mostly terrestrial network are down, the use of digipeater enable to convey information from one location to another location
- **<u>Mission Objectives</u>**
  - To connect Amateur Radio Operators all around the world
  - To demonstrate the Software Digipeater in Space
  - To  provide early warning at the time of disaster to the nearest ground station if satellite is maintaining Line of Sight (LoS) with the operating station

# Requirements (DP-M)

| ID | System Requirements | Design Requirement | Verification Requirements |
|---|---|---|---|
| **SR1.1** | **Digipeater mode** of S2S shall utilize the COM-BUS-Subsystem | Digipeater **packet format** shall be similar to the COM-BUS-Subsystem but **packet length** should be **smaller** in size | Check the size of packet while transmitting and receiving along with packet format |
| | | Data rate shall be low to meet the full success level | Shall ensure **1200 bps** data rate |

# Requirements (DP-M)

| ID | System Requirements | Design Requirement | Verification Requirements |
|---|---|---|---|
| **SR1.1** | **Digipeater mode** of S2S shall utilize the COM-BUS-Subsystem | Shall use **low-power GFSK** Scheme at **Amateur Radio UHF Band** (**435.375MHz** for uplink, **437.375MHz** for downlink) | Shall ensure the UHF GFSK signal **transmitting power** is at **+14 dbm** |
| | | Digipeater mode shall use **Power Amplifier** for transmitting | Shall ensure **transmitting power** is at **+32dbm** |

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slippers2sat-2

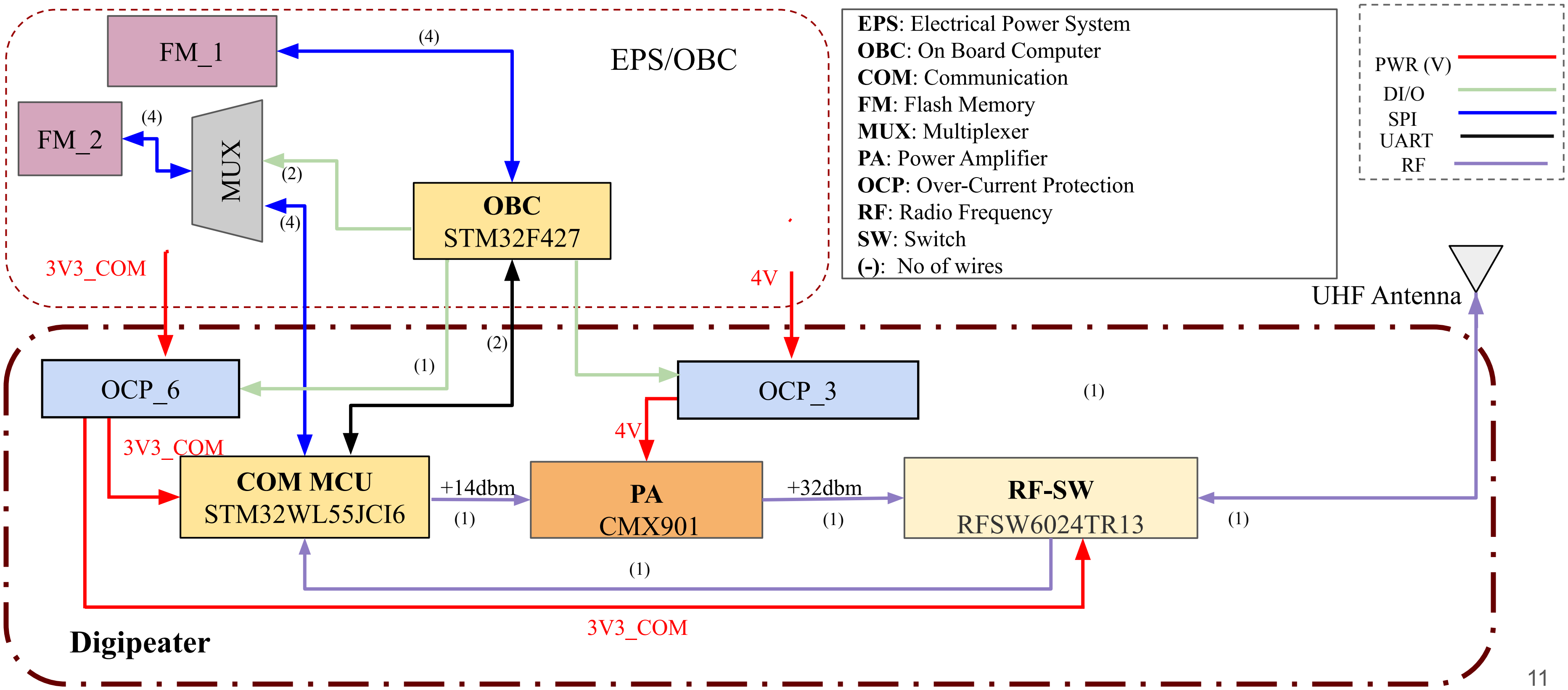
# Requirements (DP-M)

| ID | System Requirements | Design Requirement | Verification Requirements |
|---|---|---|---|
| **SR1.1** | **Digipeater mode** of S2S shall utilize the COM-BUS-Subsystem | Received data shall be stored to memory (facillate **S&F** operation) | Shall forward received data to OBC through **UART** (9600 baud rate) and store to FM_2 |
| | | Stored data shall be re-transmitted (facillate **S&F** operation) | Received Data shall be transmitted using downlink frequency |

# Requirements (DP-M)

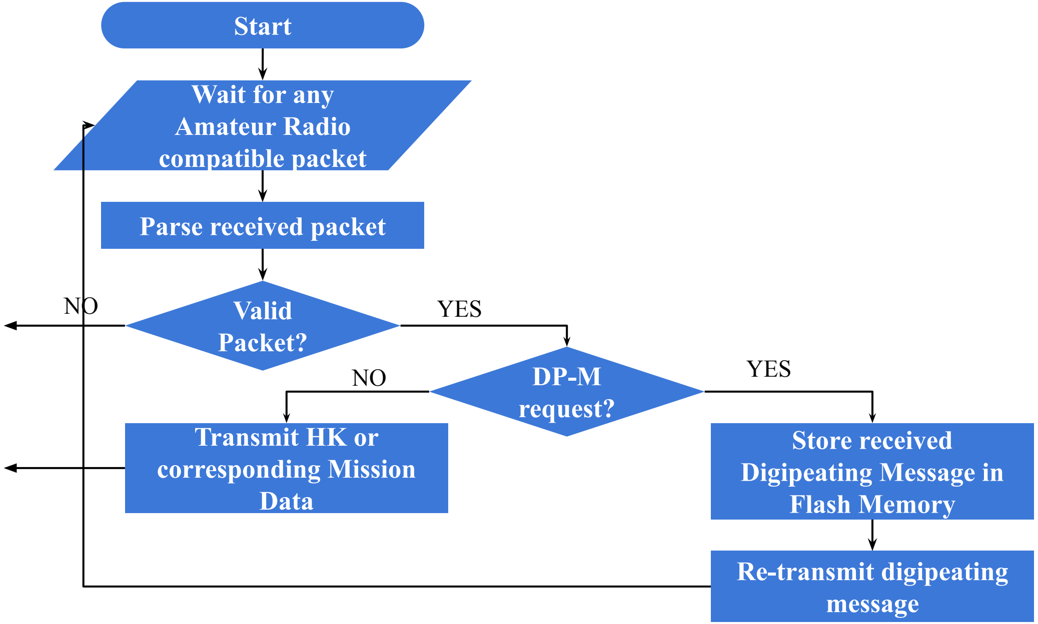
| ID | System Requirements | Design Requirement | Verification Requirements |
|---|---|---|---|
| **SR1.2** | Digipeater shall be operated all the time in orbit | Shall **automatically start** to operate when **digipeating message is received** and meets the **power requirements (Power Budget = 1290.090mWh)** | Shall define **maximum time (5520 sec)** to operate in 1 orbit |
| | | Shall **automatically stop** to operate after storing and re-transmitting of received data | Shall define power budget **threshold (1136.66mWh)** to operate Digipeater Mission |

# Block Diagram




**EPS**: Electrical Power System
**OBC**: On Board Computer
**COM**: Communication
**FM**: Flash Memory
**MUX**: Multiplexer
**PA**: Power Amplifier
**OCP**: Over-Current Protection
**RF**: Radio Frequency
**SW**: Switch
**(-)**: No of wires

# Digipeater Flow Chart



# Feasibility Test (Size)

| Mission Name | Components | Volume | Compatibility |
| --- | --- | --- | --- |
| **Digipeater (MSN1)** | NXT_GEN_CUBUS | $50 \times 100 \times 113 mm^3$ | YES |
| **CAM (MSN2)** | Imaging Sensor | $6 \times 6 \times 0.75\ mm^3$ | YES |
| | Lens x2 | $[\pi \times (15)^2 \times (22.5) mm^3$ $(\pi r^2 h)] * 2$ | YES |
| **ADCS (MSN3)** | Magnetorquer x2 | $72 \times 15 \times 14 mm^3$ | YES |
| | GPS Receiver | $25.14 \times 25.13 \times 4.4 mm^3$ | YES |
| **EPDM (MSN4)** | Magnetometer x4 | $25.4 \times 25.4 \times 7.1 mm^3$ | YES |

# Feasibility Test (Weight)

| Mission Name | Components | Mass | Compatibility |
|---|---|---|---|
| **Digipeater (MSN1)** | NXT_GEN_CUBUS | ~0.476kgs | YES |
| CAM (MSN2) | Camera Sensor | 0.017kgs | YES |
| | Lens | 0.09kgs | YES |
| ADCS (MSN3) | Magnetorquer x2 | 0.03kgs | YES |
| | GPS Receiver | 0.18kgs | YES |
| EPDM (MSN4) | Magnetometer x4 | 0.059kgs | YES |

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# Feasibility Test (Power)

**3SP_E(30%) = 1173.6 mWh** **3SP_E(32%) = 1291.2 mWh**

| BUS/ Mission | | Telemetry | | Mission | | | |
|---|---|---|---|---|---|---|---|
| | Burner | Beacon | HK-Data | DP-M | ADCS | CAM | EPDM |
| EPS/OBC | 1136.66 | 1136.66 | 1136.66 | **1136.66** | 1136.66 | 1136.66 | 1136.66 |
| COM TX (Beacon) | OFF | 49.50430556 | OFF | OFF | OFF | OFF | OFF |
| COM TX (Downlink) | OFF | OFF | 129.1416667 | OFF | OFF | OFF | OFF |
| COM RX (Uplink) | OFF | 24.288 | 24.288 | OFF | 24.288 | 24.288 | 24.288 |
| DP-M TX (Downlink) | OFF | OFF | OFF | **129.1416667** | OFF | OFF | OFF |
| DP-M RX (Uplink) | OFF | OFF | OFF | **24.288** | OFF | OFF | OFF |
| ADCS | OFF | OFF | OFF | OFF | 1.739444444 | 1.739444444 | OFF |
| CAM | OFF | OFF | OFF | OFF | OFF | 25.10866667 | OFF |
| EPDM | OFF | OFF | OFF | OFF | OFF | OFF | 130.35 |
| Burner Circuit | 33 | OFF | OFF | OFF | OFF | OFF | OFF |
| GPS | OFF | 0.092 | 0.092 | **0.092** | 0.092 | 0.092 | 0.092 |
| Total Energy (mWh) | 1169.660 | 1210.452 | 1290.090 | **1290.090** | 1163.196 | 1187.796 | 1291.390 |

# Feasibility Test (Cost and Logistics)

- Digipeater is NXT_GEN_BUS driven mission and it meets the size, weight and power requirement for 1U CubeSat
- Digipeater is **software-based** and functionality can be modified logically

# Success Criteria

| Digipeater Mission | Success Criteria |
|---|---|
| Minimum Success | Satellite should receive the digipeating short message from ground station and retransmit same message to other amateur radio station |
| Full Success | Satellite should operate digipeater till EOL (End of Life) |